2.2.3 SCA Normal Mode Measurements

This exercise is to help you determine the relationship between the upper level discriminator (ULD) and LLD dial settings.

- Set the SCA to *normal* mode
- For various LLD and ULD settings find the minimum and maximum pulse amplitudes that result in logic pulse output from the SCA.

3 Personal Computer Operation and the Multichannel Analyzer (MCA)

The multichannel analyzers you will use throughout the remainder of the term are installed in an expansion slot of you lab station computer. With the aid of the instructor, familiarize yourself with ORTEC Maestro, the software used to produce and display a multichannel spectrum. The input pulses from the amplifier to the MCA must be positive in polarity between 0-10 V.

3.1 Equipment Setup

3.1.1 Measurement Setup

- Connect the pulser to the amplifier
- Connect the output of the amplifier to the input of the MCA (on the back of your computer)
- Set the pulser amplitude and amplifier gain to produce pulses that meet the MCA input requirements
- NOTE: Since the pulser amplitude is fixed, all pulses should be stored in a single channel

*What channel is associated with the pulser amplitude?
*How many channels does your MCA have?

3.2 MCA Measurements

3.2.1 MCA General Measurements

- Set the pulser frequency to about 1 kHz
- Split the signal from the amplifier, directing one branch into the oscilloscope and the other into the MCA
- Record the MCA channel and pulse amplitude while varying the pulser amplitude. Do not erase the MCA after each step. The result is a series of spikes that mark the channel corresponding to each discrete pulser amplitude.
- Record the net count under each peak measured for the same live time.
- Change the amplifier gain, and repeat the measurement.

Example table for recording data

Pulser amplitude	Amplifer gain	Amplifer amplitude	Channel number	Channel counts
Amp 1	gain 1			
Amp 2	gain 1			
...Amp 5	gain 1			
Amp 1	gain 2			
Amp 2	gain 2			
...Amp 5	gain 2			
Amp 1	gain 3			
Amp 2	gain 3			
...Amp 5	gain 3			

3.2.2 MCA Dead Time Measurements

- After every pulse which the MCA accepts and analyzes, it is 'dead' for a certain amount of time, i.e., it will not accept additional pulses during that time interval. The pulser can be used to measure the MCA dead time by using the double-pulse feature.

- PULSER: Set the rise time to be short
- PULSER: Set the fall time to be 10 μs.
- AMPLIFIER: Set the shaping time to 0.5 μs.
- Vary the pulse amplitude until the recorded pulses lie somewhere near the middle of the MCA channel range
- PULSER: Switch to double-pulse mode and observe the amplifier output on the scope while also supplying it to the MCA
- NOTE: The spacing between double-pulses can be varied using the 'delay' control on the pulser
- Start with the minimum delay on the pulser (or 10 μs).
- Slowly increase the delay until the MCA begins recording a distinct second peak corresponding to the smaller, delayed second pulse (in a channel far removed from the first one)
- NOTE: The double pulse feature of the pulser produces two pulses of the same amplitude. To produce pulses of differing amplitudes that can be resolved with the MCA, the pulser output must be shaped by the amplifier and then passed to the o-scope.
- Measure the corresponding time spacing on the scope which is now the corresponding MCA dead time. Repeat using several different initial pulse amplitudes and record the MCA dead time and the initial pulse channel number.

4 Required Plots

Create the following plots using your computer and the data you measured. Include them in your lab report.

Section 2.2.1

- Plot the SCA input amplitude vs. the corresponding LLD setting.
- Plot the SCA input amplitude vs. the rise time of the SCA input signal.

Section 2.2.2

- Plot the range of input pulse heights vs. the window width to check for linearity (plot all three LLD settings on the same graph).

Section 2.2.3

- Make necessary plots between appropriate variables based on your best judgement.

Section 3.2.1

- Plot the amplifer output amplitude vs. MCA channel number and net counts for three different amplifer gains.

Section 3.2.2

- Plot the MCA dead time vs. initial pulse channel number.

5 Post-Lab Questions

1. Why is integral linearity important?
2. How many SCAs would you need to equal the discriminatory capability of your specific MCA?
3. What are the advantages and disadvantages of increasing the number of channels?
4. Based on your results from the plot in section 3.2.2, what type of ADC is in your MCA?
5. Explain ballistic deficit and pulse pileup. Can you minimize the problems associated with both of these effects at the same time? Why or why not?